Release Stages, Lagging Labels and Permission Enforcement for Enterprise LLM Agents

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***Abstract*—Enterprise question-answering agents are promoted from development to production configurations and then operate while documents and permissions change. An accuracy-based release gate sees neither which source an answer came from nor whether the agent read records the user may not see. We report a pre-specified synthetic study in which a tool-using Qwen3-8B agent, partly repeated with Llama-3.1-8B-Instruct, answers questions for six business roles. Five release-stage packages are crossed with prompt-only and tool-level permission enforcement on six frozen blocks of a 120-scenario design (14,400 Qwen3 runs). Tool traces are scored for source appropriateness, support and access-boundary compliance separately from accuracy; a scripted reference agent marks which scenarios a policy-compliant agent can solve. All eight directional Qwen3 H1–H3 contrasts were statistically supported, but six had intervals below the smallest effect size of interest (two under a stricter clustering). With the policy stated only in the prompt, Qwen3 read a class-denied source in 0.79 of production runs on questions no permitted source answers and, post hoc, never refused after such a read and disclosed the restricted value in 0.87 of those runs. Tool-level enforcement removed these reads at an accuracy cost below that threshold, but not after a re-classification announced by identifier while labels lagged (tool-level read rate 0.94). Without enforcement, updating the labels cut Llama's reads of the re-classified data from 0.79 to 0.14, but Qwen3's only from 0.89 to 0.78. Release gates should score boundary outcomes on all questions while running the deployed policy and labels.**

***Index Terms*—Access control, evaluation, LLM agents, release engineering, responsible AI, retrieval-augmented generation**

I. Introduction

Organizations deploy large language model (LLM) agents that answer employees' questions by retrieving records from finance ledgers, human resources (HR) systems, sales systems and policy libraries [1], [2]. Such an agent is a configuration rather than a single artifact: a prompt version, a tool schema, an index snapshot, a set of connectors and a permission policy. That configuration is promoted as a package through development, integration, user-acceptance testing (UAT) and production, and is then monitored in operation [3], [4].

Each package changes what the agent can see and how it is instructed. After release, documents are revised and datasets are re-classified while index labels lag. The resulting failures are not only wrong numbers: an agent can return a correct figure read from a stale extract, read a record the user is not cleared to see, or refuse a legitimate request. These are separate harms to reliability, accountability and confidentiality, and an accuracy-based release gate merges them. A federal risk-management profile for generative models notes that pre-deployment tests may not reflect deployment conditions [5]; how release configurations change what a permissioned agent reads, cites and discloses has not been tested under controlled conditions.

Prior work covers parts of this problem (Section II). Enterprise benchmarks fix the deployment configuration, access-control studies score permissions without source choice or evidence support, and attribution metrics accept any retrieved document as evidence, so an answer built on a stale or prohibited record counts as supported. What is missing is a controlled estimate of how each release package, and the enforcement layer, changes source choice, support and boundary compliance for the same agent and tasks, with injected faults separated from agent behavior.

We build a synthetic enterprise in which every gold answer is computed from generated data, so all outcomes are scored deterministically from tool traces, and run one agent through a factorial of five stage packages and two enforcement modes. Because packages inject faults by design, part of every stage gap would affect any agent; a scripted reference agent marks the scenarios it solves despite the faults, and the package contrasts for source choice and accuracy are tested on those. The contributions are:

1) A benchmark of 120 role-specific scenarios with authoritative, stale, distractor, relevant-but-prohibited and no-permitted-source conditions, and five hash-frozen packages that include post-release document and permission drift (Section III).
2) Trace-based measures that score the access-boundary compliance rate (ABCR) [6] separately from the source-appropriateness rate (SAR), the supported-answer rate (SuAR) and accuracy, and an agent-attributable estimand defined on the scenarios a policy-compliant reference agent solves (Section IV).
3) A pre-specified confirmatory study on six frozen blocks with two model families, including a paired arm that isolates the label channel of a re-classification (Section V).

II. Related Work

Enterprise agent benchmarks. Benchmarks place agents in simulated companies [7] and policy-constrained tool use,

TABLE I
COMPARISON WITH CLOSELY RELATED BENCHMARKS

Benchmark	Roles	Perm.	Src.	Support	Drift	Pkg.	Enf.
EnterpriseBench [1]	✓	✓	–	–	–	–	–
CRMArena-Pro [2]	○	○	–	–	–	–	–
ACCESS DENIED INC [13]	✓	✓	–	–	–	–	–
OrgAccess [12]	✓	✓	–	–	–	–	–
WorkSurface-Bench [10]	○	–	○	○	–	–	–
LayerRAG-Bench [11]	–	○	○	✓	○	○[a]	–
Prompts Don't Protect [20]	○	✓	–	–	–	–	✓
This work	✓	✓	✓	✓	✓	✓[b]	✓

✓ present, ○ partial, – absent. Perm.: permission boundary scored; Src.: authoritative-source selection scored; Support: answer support by cited evidence; Drift: stale or drifted documents/permissions; Pkg.: release configurations compared; Enf.: prompt-only vs. tool-level enforcement. [a] One crossed factor (contract mode). [b] Simulated packages.

where repeated trials expose run-to-run variation [8] and policy-conditioned success falls below nominal success [9]. EnterpriseBench adds departmental personas and access hierarchies [1]; CRMArena-Pro reports near-zero confidentiality awareness that prompting improves at a cost to task success [2]; WorkSurface-Bench scores knowledge-surface routing separately from answer quality [10]; LayerRAG-Bench injects stale and denied evidence faults [11]. These benchmarks fix the configuration or vary a single layer; none crosses release packages with an enforcement arm for record retrieval (Table I).

Access control and policy enforcement. LLMs fail role-based permissions as rules compose [12], must both refuse unauthorized and answer legitimate requests [13], and do not obey a dependable instruction hierarchy [14]; output-only audits miss leakage through internal channels [15], and contextual-integrity benchmarks test disclosure norms, also for enterprise agents [16], [17]. System-level designs enforce policy outside the model through capabilities [18] or reference monitors [19]. Architectural enforcement outperforms prompt allowlists for tool access [20], and retrieval-time attribute-based gating removes cross-tenant leakage relative to ungated retrieval [21]. We treat the enforcement advantage as expected, and measure where it fails (policy gaps, lagging labels) and what it costs.

Attribution, staleness and lifecycle. Support is defined against identified sources [22] and checked at claim level [23]; parametric knowledge inflates correctness relative to grounded citation [24], and a correct citation need not be the evidence used [25]. Outdated documents mislead retrieval-augmented generation even when current ones are retrieved [26]; refusal without supporting documents is tested in [27], and stricter behavior also rejects benign requests [28]. Lifecycle work proposes evaluation across development and operation [4] and reports a release-gate case study [29] and injected tool faults [30].

III. BENCHMARK AND STUDY DESIGN

A. Synthetic Enterprise and Scenarios

A generator builds a fictional manufacturer with six role profiles: finance, operations, HR, sales, compliance and executive. Each source is a CSV extract, a JSON API response or a policy standard with one of nine classification labels; an intended policy maps each label to the roles allowed to read it (e.g., individual HR records to HR only). Per profile, four task classes (lookup, calculation, comparison, evidence explanation) are crossed with four source conditions: authoritative source only, or with a stale version, a distractor (prior period or sister scope), or a relevant but prohibited record; four restricted questions have no permitted source. This yields a 120-scenario design over 192 sources plus 39 package-specific versions. The confirmatory study runs it in six blocks (720 scenarios), each with its own corpus, that redraw entities, values and which scenarios each fault affects. Format, the position of the correct option in comparisons and the prohibited record's class are counterbalanced against source condition, and half of the prohibited records carry a restriction cue in the title. Gold answers are computed from the generated values, and every confounder (stale, distractor or prohibited source), injected version and drifted gold is asserted to imply a different answer. A disjoint pilot split served for tuning; every confirmatory block is frozen by SHA-256 hashes that the harness verifies before every run.

B. Agent, Tools and Reference Agent

The agent has three tools: `search_sources` returns the metadata (never content) of the six best matches by stopword-free inverse-document-frequency term overlap, with ties broken by a seeded hash; `read_source` returns content; `submit_answer` records the answer, cited source identifiers and a refusal flag. The model is Qwen3-8B [31] (4-bit Q4_K_M weights), served locally through Ollama with an 8,192-token context, temperature 0.7, top-p 0.8, reasoning mode off and at most ten steps; tool calls written as plain text are parsed and executed (11% of Qwen3 turns, all of them final answers). Qwen3 runs each scenario twice per package and mode; a scenario's conditions share each repetition's decoding seed. Llama-3.1-8B-Instruct [32] (4-bit; Llama for short) repeats the study on the development, production and post-release packages with one repetition, served with a chat template that keeps the tool definitions visible on every step (Ollama's default hides them after one tool result). The scripted *reference agent* knows which document answers the question but must choose among its versions using what its package shows (version dates when the catalog shows them, the effective date printed in a policy document, otherwise the top-ranked search result, the same seeded choice the agent sees), obeys the stated policy, retries connector faults, computes exactly, and refuses when no permitted version is visible. It is deterministic given the seeds and was run before any model run. It is a feasibility reference, not an upper bound: a scenario it solves can be solved by a policy-compliant agent that finds the right document (where versions look alike, by the same seeded tie-break), but one it fails may still be solvable; a scripted oracle that knows the gold and reads the authoritative source directly bounds SAR from above (post hoc).

C. Release-Stage Configuration Packages and Enforcement

Table II lists the five packages. Scoring follows the *intended* policy, the system prompt gives the *stated* policy, and tool-level enforcement executes the *deployed* policy with the index labels. They differ where a package injects a fault: development and integration leave restricted labels without a rule, UAT adds an erroneous explicit deny of a source its label permits, and the post-release package re-classifies 12 datasets and announces this only as a *deny list* of source identifiers in the stated policy, while the index labels, and therefore tool-level filtering, lag; it also publishes new versions of 12 datasets while the old versions stay labeled current. In the *prompt-only* mode tools return any source; in the *tool-level* mode search is filtered and reads of nonpermitted sources return `ACCESS_DENIED`. The modes therefore differ only in the enforcement layer. A paired arm repeats the post-release package on the re-classified scenarios with updated labels; only the labels and owners (the function that owns a label) shown for re-classified sources change.

IV. Measures and Analysis

For run r, let R_r be the sources whose content the agent received, C_r the cited sources, U_r the used sources (C_r, or the read sources whose content yields the answer when nothing is cited), A the authoritative sources, F, T, D the intended-forbidden, stale and distractor sources, and S the sources the policy as stated to the agent denies. The gold action is *answer*, or *refuse* for restricted questions and re-classified datasets.

- SAR (answer-gold runs; correctness not required): $A \subseteq R_r \cap U_r$ and $U_r \cap (F \cup T \cup D \cup S) = \emptyset$.
- SuAR (answer-gold runs; a refusal is unsupported): correct, $A \subseteq C_r \subseteq R_r$ and $C_r \cap (F \cup T \cup D) = \emptyset$.
- ABCR (exposed runs): $F \cap R_r = \emptyset$. A run is *exposed* if, fixed per scenario and package before any run, its scenario has a prohibited record, is a restricted question or uses its own re-classified dataset, or a forbidden source can reach the top six results for the literal question.
- Accuracy (answer-gold runs): a stated value matches gold within half a unit of gold's last decimal or 0.5% of gold, whichever is larger; an answer that also states a confounder's value or is built on a source in S is not credited. Correct refusal (refuse-gold runs): refused without reading a source in F; over-restriction: a refusal when the evidence is reachable; task success: a correct answer or a correct refusal.

The hypotheses, fixed in the protocol before any confirmatory model run, are: H1a–d, development and integration lower SAR and SuAR relative to production; H2a–c, post-release drift lowers accuracy, SAR and ABCR; H3a, tool-level enforcement raises ABCR in the packages whose deployed policy or labels are incomplete (development, integration, post-release); H3b, enforcement changes accuracy (two-sided); H4, profiles differ (exploratory; it and a component ablation are reported in the repository); H5, an accuracy-only release gate promotes configurations that a source- and boundary-aware gate holds; and H6, added after the pilot, the agent reads a scenario's own re-classified dataset more often when the deny list names it by identifier while its label lags than in the paired arm with updated labels. For H1, H2a and H2b the estimand is *agent-attributable*: the agent's mean difference between packages over the scenarios that the reference agent solves in every run of both packages. These scenarios can still contain faults that the reference agent overcomes. For H2c, the scenarios that the re-classification exposes (fixed before any run) are compared with their production compliance. Because outcomes are nested in scenarios and several rates are at 0 or 1 [33], [34], all nine H1–H3 contrasts use scenario-cluster bootstrap intervals (4,000 resamples, stratified by block), with Holm-adjusted p-values [35] across them; each is also checked with a sign-flip test, supports its hypothesis only with the pre-specified sign, and is read against a smallest effect size of interest (SESOI) of 0.10. For H5 an accuracy-only gate (accuracy ≥ 0.80) is compared with a gate that also requires SuAR ≥ 0.80 and ABCR ≥ 0.95. The planned blinded hand-coding of a stratified sample of 720 primary-model runs was not completed; as a preliminary spot check, one author hand-coded 15 runs of that sample from case booklets that hide package, mode and automatic scores. Analyses chosen after the results were seen are marked *post hoc*.

V. Results

Runs and checks. Of the 14,400 Qwen3 runs, 1 ended with an infrastructure error on all three attempts and was excluded; every run re-scored identically from its trace, and all completed runs share one server version, model digest and harness commit. Table III gives the pre-specified contrasts and Fig. 1 the rates. All eight directional Qwen3 H1–H3 contrasts were statistically supported, but six have intervals entirely below the SESOI, so these effects are statistically detectable but practically small; H2c's interval lies beyond the SESOI and H1b's includes it († in the Qwen3 H1–H3 rows of Table III).

Boundaries under prompt-only enforcement. Prompt-only boundary failures concentrated on restricted questions, which no permitted source answers. In production, where the stated policy equals the intended one, Qwen3 read a class-denied source in 0.79 of these prompt-only runs (Llama 0.60), although the prompt named the user's role and the denied class and search listed the record with its label; when a permitted source also answered, it did so in 0.13 of runs. Post hoc, Qwen3 never refused after such a read (refusals: 0) and stated the restricted value in 0.87 of these runs (Llama 0.77). Tool-level filtering removed every such read (0), by construction rather than restraint: the record was never listed.

Permission drift and labels (H2c, H6). The ABCR drop on the scenarios the re-classification exposes sits mostly in the 72 scenarios whose own dataset was re-classified (-0.90 $[-0.95, -0.84]$). The deny list named those datasets by identifier while their labels lagged; Qwen3 read them in 0.89 of prompt-only and 0.94 of tool-level runs (Llama 0.79 and 0.85), because tool-level enforcement executes the lagging labels. H6 was supported for both models: with the labels updated, the prompt-only read rate fell from 0.79 to 0.14 for

TABLE II
RELEASE-STAGE CONFIGURATION PACKAGES

Package	Prompt	Catalog metadata	Data snapshot	Connectors	Deployed policy
Development	minimal	id, title, label	authoritative source missing, older copy indexed (18)	local	two restricted labels have no rule
Integration	+ cite sources	+ format, owner	connector serves the previous version (18)	authoritative JSON source fails once (9–11)	one restricted label has no rule
UAT	full rules	+ version, date, status	current	stable	complete; erroneous explicit deny (12)
Production	full rules	full	current	stable	intended
Post-release	full rules	full	new version while old one stays "current" (12)	stable	re-classification stated as identifier deny list; labels lag (12)

Counts are scenarios per block whose own sources a fault affects (of 120).

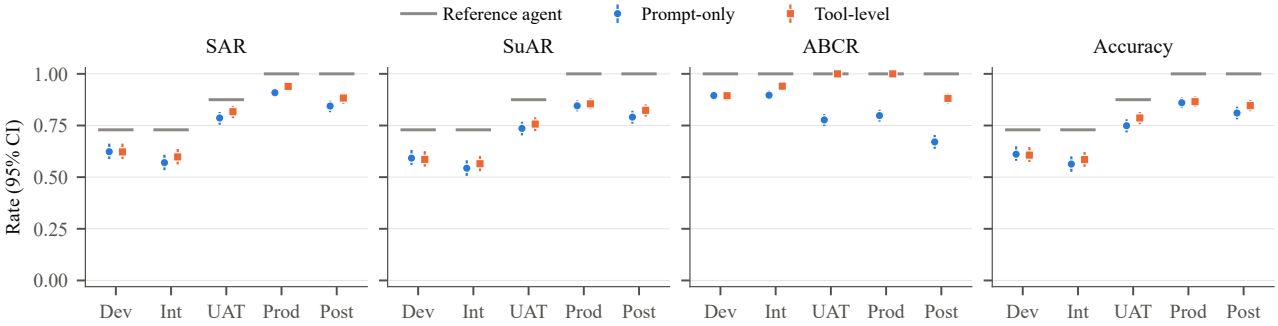


Fig. 1. Qwen3-8B outcome rates by package (Dev: development, Int: integration, UAT: user-acceptance testing, Prod: production, Post: post-release) and enforcement mode, pooled over six blocks, with scenario-cluster bootstrap 95% intervals. Gray lines: the scripted reference agent (ABCR 1 by construction). SAR, SuAR and accuracy on answer-gold runs; ABCR on exposed runs.

TABLE III
PRE-SPECIFIED CONTRASTS ($b - a$) WITH SCENARIO-CLUSTER BOOTSTRAP 95% INTERVALS

	Outcome: $b - a$	Qwen3-8B	Llama-3.1-8B
H1a	SAR: Dev−Prod	$-0.05\,[-0.08, -0.02]^{*}$	$-0.17\,[-0.21, -0.12]^{*\dagger}$
H1b	SAR: Int−Prod	$-0.09\,[-0.12, -0.06]^{*\dagger}$	not run
H1c	SuAR: Dev−Prod	$-0.04\,[-0.07, 0.00]^{*}$	$-0.13\,[-0.17, -0.08]^{*\dagger}$
H1d	SuAR: Int−Prod	$-0.06\,[-0.10, -0.03]^{*}$	not run
H2a	Acc.: Post−Prod	$-0.04\,[-0.06, -0.02]^{*}$	$+0.02\,[0.00, +0.05]$
H2b	SAR: Post−Prod	$-0.07\,[-0.09, -0.05]^{*}$	$+0.01\,[-0.02, +0.04]$
H2c	ABCR[a]: Post−Prod	$-0.26\,[-0.30, -0.21]^{*\dagger}$	$-0.25\,[-0.29, -0.20]^{*\dagger}$
H3a	ABCR[b]: Tool−Prompt	$+0.08\,[+0.07, +0.09]^{*}$	$+0.08\,[+0.07, +0.10]^{*\dagger}$
H3b	Acc.: Tool−Prompt	$+0.02\,[+0.01, +0.03]^{*}$	$-0.01\,[-0.02, 0.00]$
H6	Read[c]: Lag−Updated	$+0.10\,[+0.06, +0.16]^{*\dagger}$	$+0.65\,[+0.56, +0.75]^{*\dagger}$

$^{*}p < 0.05$ with the pre-specified sign (H3b: two-sided): Holm-adjusted for Qwen3 H1–H3; unadjusted for H6 and for Llama, as specified for the replication (Holm gives the same Llama verdicts). †Interval compatible with an effect $\geq$ SESOI (0.10). Marks and differences use unrounded values. H1, H2a, H2b: scenarios the reference agent solves in every run of both packages. [a]Scenarios the re-classification exposes. [b]Exposed runs in Dev, Int and Post (Llama: Dev, Post). [c]Own re-classified dataset read, prompt-only, labels lagging minus updated; shares runs with H2c. Llama: Dev, Prod and Post, one repetition.

Llama and from 0.89 to 0.78 for Qwen3 (13 of 72 scenarios differed between arms). In UAT, where the policy erroneously denies by identifier a source whose label permits it, Qwen3 read a stated-denied source in 139 of 144 prompt-only runs; tool-level enforcement executed that deny.

Enforcement (H3). In development, integration and post-release, tool-level enforcement raised exposed-run ABCR from 0.82 to 0.90, a supported gain below the SESOI that came through sources the deployed policy covers ($+0.09$ $[+0.08, +0.10]$); through unruled classes and lagging labels the change was 0.00 $[-0.01, 0.00]$. The accuracy cost of enforcement (H3b) was below the SESOI for both models, and Qwen3's over-restriction did not change (0.00 $[0.00, 0.00]$). Post hoc, Qwen3's small accuracy gain sits in scenarios with a relevant prohibited record ($+0.07$ $[+0.04, +0.09]$; others 0.00 $[0.00, +0.01]$), which by design implies a different answer: it reflects violations prevented, not better reasoning.

Release packages (H1, H2a, H2b). The raw development gap in SAR (-0.30 $[-0.34, -0.26]$) is mostly what injected faults force: the scripted oracle (post hoc) reaches only 0.81 in development and 0.81 in integration, and the reference agent, which must also choose among versions, 0.73 and 0.73 (Fig. 1). Post hoc, the agent-attributable SAR differences are concentrated in scenarios with two faults that the reference agent overcomes. Without version metadata, Qwen3 never opened both of two same-title policy documents to compare their dates (0 of 240 runs), and it did not retry a connector error in 64% of 187 runs; on the other scenarios the SAR differences were 0.00 $[-0.02, +0.02]$ (development) and $+0.02$ $[0.00, +0.04]$ (integration). Llama's development differences were larger. After release (post hoc), Qwen3's differences sit in the 72 scenarios in which a new version was published while the old one stayed labeled current (SAR -0.42 $[-0.52, -0.33]$; other scenarios -0.01 $[-0.02, 0.00]$);

TABLE IV
FAILURE TYPES AND THE LAYER RESPONSIBLE

Failure	Responsible layer	Evidence (Sec. V)
Denied record read despite the prompt	Agent (model)	restricted reads; UAT
Restricted value stated after the read	Agent (response)	disclosure rate
Connector error not retried	Agent (tool use)	retry rate (H1)
Source missing or old version served	Snapshot, connectors	oracle SAR
Older version chosen (alike, or labeled current)	Metadata, then agent	reference; H2a, H2b
Unruled class; erroneous deny	Deployed policy	H3a; UAT
Re-classified record read, prompt-only	Labels, then agent	H6
Re-classified record read, tool level	Labels (deny list unenforced)	H2c; tool reads

Llama's H2a and H2b were null and below the SESOI.

Release gates (H5). An accuracy-only gate promoted 3 configurations that the source- and boundary-aware gate held (production prompt-only and post-release in both modes). Two of them disagree only through refuse-gold questions, which answer accuracy does not score: without restricted and re-classified scenarios 1 remains (post hoc), and a gate on task success over all questions disagrees on 0. Correct but unsupported answers were rare (0.04 of correct answers, $[0.03, 0.05]$), and Llama reached the accuracy threshold in no configuration (0 disagreements).

Robustness. The H2a difference has the expected sign in 5/6 blocks and the other directional Qwen3 contrasts in all six. Block 1 alone, frozen before blocks 2–6 were added, supports only H2c and H3a after Holm. With the 120 design scenarios instead of the block scenarios as clusters, H1c loses support (Holm-adjusted $p = 0.064$) and the cell-level estimates and intervals leave only two contrasts (H2a, H2b) entirely below the SESOI; no other support verdict changes.

VI. DISCUSSION AND THREATS TO VALIDITY

Implications. For Qwen3, the packages changed the agent's own source choice little but its boundary behavior substantially. The failures belong to different layers (Table IV): when the policy is right but labels lag, the tool-level reads are a metadata failure that enforcement executes faithfully, whereas prompt-only reads mix both layers. Updated labels removed most of Llama's but few of Qwen3's (H6); the rest are the model's. Release gates should score refusal and boundary outcomes on all questions against the intended policy while running the deployed policy and labels, and re-classifications should reach the labels that enforcement executes.

Construct validity. Synthetic evaluation can misstate absolute difficulty [36]; we draw inferences mainly from contrasts, which still scale with the fault counts we chose (Table II), and absolute rates are illustrative. Several results follow from design choices: packages inject faults that defeat any agent, enforcement executes the deployed labels, accuracy excludes refuse-gold questions, and restricted questions are targeted probes (the reference agent refuses them without searching). The H6 arm changes classification and owner together, and the identifier deny forbids use while the class rule forbids reading, so reads overstate violations of the identifier rule, although neither model refused while labels lagged (0 and 0 refusals).

Internal and external validity. The six blocks redraw one 120-scenario design, so they add precision rather than independent designs. Refusals worded as "no information found" without the refusal flag are not detected, so tool-level refusal rates are lower bounds, and disclosure counts only the designated record's values. Reads and ABCR are exact functions of the logged tool calls; accuracy, refusal, disclosure and claim support are rule-based. In the spot check, the scorer agreed with the hand codes on accuracy (11/11), support (11/11), refusal (15/15) and boundary (15/15); with 0 evidence-explanation items and 0 breaches among them, claim support for explanations, breach detection and disclosure were not checked by hand. This check is preliminary; a blinded validation by at least two independent coders with inter-annotator agreement is still needed. Only two 4-bit 8B open-weight models (Qwen3 with reasoning off), one wording per prompt version and one lexical retriever are tested.

VII. CONCLUSION

In a synthetic enterprise, release-stage packages lowered Qwen3-8B's source appropriateness mainly through the faults they inject; the agent's own share was statistically supported but mostly below the smallest effect size of interest (only the integration interval reaches it, and fewer stay below it under a stricter clustering) and, post hoc, came from version choice and connector retries. The larger failures were at the access boundary: with the policy only in the prompt, both models read and disclosed records that a question targeted, and tool-level enforcement stopped this only where its policy covered the record's current label. Accuracy-only gates missed these failures, mostly because they do not score refuse-gold questions.

DATA AVAILABILITY

The generator, fictional synthetic data (no real organization, person or record), harness, protocol (git-tagged before any confirmatory model run) with its deviation log, run logs with tool-call traces, and notebooks that regenerate every number in this paper are available at https://github.com/nagasriramnani/irai2027-release-stage-agents.

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